

FRONTEND (React, JS, TS, State Management)

1 What is React?

Answer:

React is a JavaScript library used to build user interfaces.

It is component-based and uses a Virtual DOM to update UI efficiently.

2 What is Virtual DOM?

Answer:

Virtual DOM is a copy of the real DOM.

When state changes:

1. React updates Virtual DOM
 2. Compares it with previous version (diffing)
 3. Updates only changed parts in real DOM
- ☞ This makes React fast.
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3 What are Hooks in React?

Hooks allow functional components to use state and lifecycle features.

Common hooks:

- `useState` → manage state
 - `useEffect` → side effects (API calls)
 - `useContext` → access global data
 - `useRef` → access DOM directly
-

4 What is `useEffect`?

Answer:

Used to handle:

- API calls

- Subscriptions
- Timers

Example:

```
useEffect(() => {
  fetchData();
}, []);
```

[] means run only once (componentDidMount).

5 What is difference between Redux and Context API?

Redux	Context API
Used for large apps	Used for small-medium apps
Has middleware	No middleware
More scalable	Simple
Boilerplate more	Less code

6 What is Zustand?

Answer:

Zustand is a lightweight state management library.
It is simpler than Redux and requires less boilerplate.

7 What is TypeScript?

Answer:

TypeScript is JavaScript with types.

Example:

```
let name: string = "Keerthana";
```

Benefits:

- Type safety
- Fewer runtime errors
- Better IntelliSense

8 What is difference between var, let, const?

var	let	const
Function scoped	Block scoped	Block scoped
Can redeclare	Cannot redeclare	Cannot redeclare
Can reassign	Can reassign	Cannot reassign

9 What is event loop in JavaScript?

Answer:

JS is single-threaded.

Event loop handles async operations like:

- setTimeout
 - Promises
 - API calls
-

● BACKEND (Python – FastAPI / Django / Flask)

10 What is FastAPI?

Answer:

FastAPI is a modern Python web framework.

- Very fast
 - Uses async
 - Automatic API docs (Swagger)
 - Uses Pydantic for validation
-

11 Difference between Django and Flask?

Django	Flask
Full framework	Lightweight
Has ORM	No built-in ORM
Admin panel included	No admin

Django	Flask
Best for large apps	Best for small apps

12 What is REST API?

REST API allows communication between frontend and backend using:

- GET → fetch
 - POST → create
 - PUT → update
 - DELETE → delete
-

13 What is Middleware?

Middleware runs before request reaches route.

Example:

- Authentication
 - Logging
 - Error handling
-

14 What is JWT Authentication?

Answer:

JWT (JSON Web Token) is used for secure authentication.

Flow:

1. User logs in
 2. Backend generates token
 3. Token stored in frontend
 4. Token sent in headers for protected APIs
-

15 What is Microservices?

Instead of one big application:

- Break into small services

- Each service handles one functionality
Example:
 - User service
 - Payment service
 - Course service
-

● DATABASE

16 What is indexing in PostgreSQL?

Index improves query speed.

Without index → Full table scan

With index → Faster search

Example:

```
CREATE INDEX idx_email ON users(email);
```

17 What is normalization?

Organizing tables to remove redundancy.

Example:

Instead of storing user address repeatedly,
create separate address table.

18 Difference between SQL and MongoDB?

PostgreSQL	MongoDB
Relational	NoSQL
Tables	Collections
Fixed schema	Flexible schema
SQL queries	JSON queries

19 What is aggregation in MongoDB?

Used to process data.

Example:

```
db.orders.aggregate([
  { $group: { _id: "$status", total: { $sum: 1 } } }
])
```

● DEVOPS & TOOLS

20 What is Docker?

Docker is used to containerize applications.

It ensures:

- Same environment everywhere
 - Easy deployment
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21 What is difference between image and container?

Image	Container
Blueprint	Running instance
Static	Running

22 What is Git rebase vs merge?

Merge → Keeps history

Rebase → Cleaner history (linear)

● SCENARIO QUESTIONS (IMPORTANT)

◆ How do you optimize React performance?

Answer:

- Use React.memo
 - Use useMemo / useCallback
 - Avoid unnecessary re-renders
 - Lazy loading
-

◆ How do you handle large data in backend?

Answer:

- Pagination
 - Indexing
 - Query optimization
 - Caching
-

◆ How do you secure APIs?

Answer:

- JWT authentication
- Role-based access
- HTTPS
- Input validation
- Rate limiting

1 What is Git?

Answer:

Git is a distributed version control system used to track code changes and collaborate with team members.

Each developer has a local copy of the repository.

2 What is the difference between Git and GitHub?

Git	GitHub
Version control tool	Hosting platform
Installed locally	Cloud platform
Tracks changes	Stores repositories

3 Important Git Commands (Must Know)

Initialize repo

```
git init
```

Clone repo

```
git clone <url>
```

Check status

```
git status
```

Add files

```
git add .
```

Commit

```
git commit -m "Added login feature"
```

Push

```
git push origin main
```

Pull

```
git pull origin main
```

```
git branch <branch-name>
```

```
git checkout <branch-name>
```

```
git merge <branch-name>
```

4 What is a Branch?

Branch is used to develop features separately without affecting main code.

Example:

```
git checkout -b feature/login
```

After completion:

```
git merge feature/login
```

5 What is Merge Conflict?

When two developers modify the same file and push changes.

Git cannot decide which change to keep.

We manually resolve conflict and commit again.

6 What is Git Rebase?

Rebase moves your branch on top of another branch.

It creates cleaner commit history.

Difference:

- Merge → Creates merge commit
 - Rebase → No extra commit (linear history)
-

7 What is Git Stash?

Used to temporarily save changes without committing.

```
git stash
git stash pop
```

Useful when switching branches quickly.

8 What is HEAD in Git?

HEAD points to the current branch or commit you are working on.

9 What is CI/CD?

CI → Continuous Integration

CD → Continuous Deployment

Whenever code is pushed:

- Automatically build
- Run tests
- Deploy to server

Example: GitHub Actions

Scenario Questions (Git)

💡 How do you undo last commit?

If not pushed:

```
git reset --soft HEAD~1
```

If already pushed:

```
git revert <commit-id>
```

💡 How do you delete a branch?

Local:

```
git branch -d branch-name
```

Remote:

```
git push origin --delete branch-name
```

💡 In your project, how did you use Git?

Best answer example:

"In my project, we followed feature-based branching. For every new feature like authentication or course module, I created a separate branch, raised a pull request, and after review, it was merged into the main branch."

This sounds professional.

DOCKER – Interview Preparation

What is Docker?

Docker is a tool used to containerize applications.

It packages:

- Code
- Dependencies
- Environment

So it runs the same everywhere.

2 Why do we use Docker?

Problem:

"It works on my machine but not on server."

Solution:

Docker ensures same environment everywhere.

3 What is a Docker Image?

Image is a blueprint/template for container.

Example:

- Python image
 - Node image
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4 What is a Docker Container?

Container is a running instance of image.

Image → Blueprint

Container → Running application

5 What is Dockerfile?

Dockerfile contains instructions to build image.

Example (FastAPI):

```
FROM python:3.10

WORKDIR /app

COPY requirements.txt .

RUN pip install -r requirements.txt

COPY . .

CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

6 Build and Run Commands

Build image:

```
docker build -t myapp .
```

Run container:

```
docker run -p 8000:8000 myapp
```

7 What is Docker Compose?

Used to run multiple services together.

Example:

- Backend
- Frontend
- PostgreSQL

Example:

```
version: '3'
services:
  backend:
    build: .
    ports:
      - "8000:8000"

  db:
    image: postgres
    environment:
      POSTGRES_PASSWORD: 1234
```

Run:

```
docker-compose up
```

8 What is Volume in Docker?

Volume is used to persist data.

Without volume → data lost when container stops

With volume → data saved

9 What is Port Mapping?

```
docker run -p 3000:3000 app
```

Left → Local machine

Right → Container